TIR

SOTA: How to explain a classifier?

Mastrorillo Noé, Tortosa Matthias, Pigeon Axel, Sinatti Antoine

Abstract

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Introduction

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

1 Perturbation model-agnostic methods

In this section, we present a set of perturbation method used in black-box models explanation. A perturbation method can be defined by the generation of perturbed samples (ex: noised features in model representation or perturbed input tensor) at the input of the black-box model. The objective is to compare the model behavior (the prediction) between a set of different perturbed inputs and the original one. Then, we can determine which one is more important for the prediction. It allows the explanation model to detail why the model took a specific decision and why. The following methods register in this type. The first one, LIME (Local Interpretable Model-Agnostic Explanation) focus on a linear approximation of local model behavior. Secondly, the Anchors method provides an amelioration of LIME by introducing the concept of IF-THEN rules to provide a better user-experience. Finally, we will briefly talk about add-ons of those methods in specific cases.

Remark (Notation). Let's introduce some notations to identify which objects we will talk about. We denote with f a black-box model we want to explain, then $f : X \longrightarrow Y$ where X is the set of instances $x \in X$ and Y the different possible predictions $f(x)$. In this section, we will talk about linear explanation model, denoted g. To identify which input representation we use, we denote by x' the interpretable representation of an input x in the black-box

model f. Then, if we construct a set of perturbed interpreted samples $\mathcal{Z}$ from x', this set contains perturbed version of x' called z'. Finally, $\Omega(g)$ express the complexity of the model g, generally defined by the number of non-zeros weights in g.

1.1 LIME (Local Interpretable Model-Agnostic Explainer)

LIME founded the principe of perturbation method for classifiers explanation [1]. It provides a local explanation of a black-box model behavior $f : \mathbb{R}^d \longrightarrow \mathbb{R}$ by training a simple explanation model g over perturbed samples. This method is *model-agnostic* because it doesn't depend on the black-box model architecture. f is considered as an oracle used to train g.

Formally, for an original interpretable representation $x' \in \{0,1\}^{d'}$ of a sample $x \in \mathbb{R}^d$, it generates a set of perturbed samples z' which are drawn by selecting uniformly some x' non-zero components. This random sampling is weighted by π_x which defines a locality around x' denoted $\mathcal{Z}$. In practical, π_x is an exponential kernel applied to a distance D. To find the optimal model g, we can define a loss function $\mathcal{L}(f, g, \pi_x)$ telling how unfaithful g is in approximating f on $\mathcal{Z}$. The objective is then to minimize $\mathcal{L}$ and having $\Omega(g)$ low enough for human comprehension. The LIME explanation is then produced by:

$$\xi(x) = \arg\min_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g) \tag{1}$$

with G the set of available explanation models [1]. Using $\mathcal{L}$ and $\Omega(g)$ to find the best model g ensures *interpretability* and *local fidelity* on the black-box model explanation. Finally, the local explanation of the model is provided by the weights w_g trained with $g(z') = w_g \cdot z'$.

As stated at beginning, LIME provides a *local* explanation of f. To ensure a global explanation, the paper proposes a Sub-modular Picks LIME (SP-LIME) which selects a set local and non-redundant LIME explanations to be given to the user [1]. To choose those explanations, the paper introduces the concept of coverage of a set of instances which defines the importance of those instances.

Definition 1.1 *Coverage*

Let g be an explanation model and $g_i = \xi(x_i)$ the explanation of the instance x_i, we construct the $|X| \times d'$ explanation matrix $\mathcal{W}$ with $\mathcal{W}_{i,j} = |w_{g_{i,j}}|$. Then, I_j is the j-column of W and denotes the global importance of that component in the explanation space. Then, the coverage of a set of instances V is defined as the total importance of the features that appear at least one instance in V:

$$c(V, \mathcal{W}, I) := \sum_{j=1}^{d'} \mathbb{1}_{\{\exists i \in V, \mathcal{W}_{i,j} > 0\}} I_j \tag{2}$$

Finally, from a maximum number of global explanations B set by the user, the pick problem is defined by:

$$Pick(\mathcal{W}, I) = \arg\max_{V, |V| \leqslant B} c(V, \mathcal{W}, I) \tag{3}$$

We want to select the highest coverage and non-redundant explanations to provide a global one.

Unfortunately, this system has limits. First, in practice, the pick problem is intractable, so a greedy algorithm is used to compute it. Secondly, LIME algorithm doesn't define clear boundaries of the explanation instances. Indeed, all black-box models cannot be locally approximated by a linear one [2]. Finally, the explanation results depend on the LIME parameters such as B, the distance metric D and the kernel π_x. Changing one of them can change the interpretation and undermining the reliability of this method [3].

References

[1] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. “"Why Should I Trust You?": Explaining the Predictions of Any Classifier”. en. In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. San Francisco California USA: ACM, Aug. 2016, pp. 1135–1144. ISBN: 978-1-4503-4232-2. DOI: 10.1145/2939672.2939778. URL: https://dl.acm.org/doi/10.1145/2939672.2939778 (visited on 02/05/2026).

[2] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. *Nothing Else Matters: Model-Agnostic Explanations By Identifying Prediction Invariance*. Nov. 2016. DOI: 10.48550/arXiv.1611.05817. URL: http://arxiv.org/abs/1611.05817 (visited on 02/21/2026).

[3] Rahel Wilking, Matthias Jakobs, and Katharina Morik. “Fooling Perturbation-Based Explainability Methods”. In: *Workshop on Trustworthy Artificial Intelligence as a part of the ECML/PKDD 22 program*. Grenoble, France, France: IRT SystemX [IRT SystemX], Sept. 2022. URL: https://hal.science/hal-03773429 (visited on 02/21/2026).